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Communicating Science

Using Stones for Bridges, Not Barriers: How Effective Dialogue Can Restore Harmony between Religion and Evolution

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Abstract

After participating in a scientific conference celebrating the 100-year anniversary of the Scopes trial and the noteworthy advancements in evolutionary science since then and then attending a nonscientific exhibit opposing evolutionary science, we came to some key observations about communication's role in the ongoing conflict between evolution and religious fundamentalism. In this article, we begin with the history of the conflict. We then highlight the research surrounding the teaching of evolution in ways that reduce conflict with Christian worldviews. Finally, we highlight the research on science communication and use this knowledge to make recommendations on how scientists can best communicate about evolution to the public so that scientists build bridges instead of barriers.

Keywords: evolution, religion, communication, cultural competence

We are a group of biologists, educators, education researchers, and student researchers in training. Much of our research has been focused on communication strategies for navigating the cultural border between evolution and a host of religious beliefs, most notably Christian fundamentalism. A series of recent shared experiences caused us to contemplate this border in a new light.

In keeping with our own advice on appropriate and effective communication strategies (which we will offer in the present article), we have chosen not to disclose the specific names of the conferences and attractions out of respect for participants and to avoid provoking identity-protective cognition that could lead to a dismissal of our message. Our commentary is on the general trend of the viewpoints expressed, not on the identity of those expressing them.

We recently had the pleasure of attending three scientific conferences in succession: a conference on communicating science effectively, a conference on the science of teaching biology, and lastly, a centennial celebration of the Scopes Monkey Trial that included a gathering of some of the greatest minds in evolution, evolution education, and the history of the evolution–religion debate within the United States. Having spent a full week interacting with experts in science, communications, and education, we made several pertinent and sometimes dissonant observations:

First, science is a powerful epistemology, that it is self-correcting in all the ways society would hope, that it can adequately explain the "unexplainable," that evolutionary theory is one of those masterful explanations, and that it is vital to our society.

Second, the majority of those individuals doing the science (the scientists)—evolutionary science, communication science, and education science—are often innovative in their ideas; thoughtful and discerning in their execution of scientific methodologies; careful, collaborative, and open for correction in their conclusions; and genuinely passionate about helping society understand the world around them (both current and historical).

And third, those doing and communicating the science, scientists, are influenced by their own lived experiences and value systems, imbued with biases, and they may inadvertently (or intentionally) communicate their own values in ways that impede effective communication to a much more heterogeneous society than the somewhat ideologically homogeneous academy in which they work and develop ideas (Stanovich 2021).

We think all can agree that the first two observations should be celebrated. But the third should warrant contemplation, especially in light of recent tensions between science and the public. In the last four decades, the world has seen an increase in political polarization across multiple issues, with the US experiencing the most dramatic affective polarization when compared with other members of the Organisation for Economic Co-operation and Development (Boxell et al. 2020). Political science researchers have not only documented this polarization but have noted new forms of divide focused less on differing ideas and more on the "repugnant" supporters of the opposing political party (Iyengar et al. 2019). We believe this contrast exists not only in the political arena but in other areas as well, such as where science may conflict with particular religious worldviews. Likewise, polling has shown a growing public mistrust of the scientific community. In just 3 years, the United States public trust in scientists has declined from 39% to 23%, and distrust increased by 15%, with prominent differences between political affiliations, further highlighting the ongoing polarization (Kennedy and Tyson 2023).

After these three conferences, we headed to two large-scale public attractions highlighting the thoughts and feelings about evolution presented from the "other side," a perspective that systematically rejects the accepted science of evolution in favor of a biblical literalist ("plain" reading), religiously oriented viewpoint,

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viewing evolutionary science as incontrovertibly irreconcilable with a true religious identity. And we made three poignant and sometimes disconcerting observations:

First, there is an underlying (and we believe, inherently charitable) motivation to these attractions driven by a genuine and passionate concern for the "souls" of the American public, from a religious perspective.

But, second, there are fundamental misunderstandings of the science of evolution that lead to erroneous conclusions about conflict among those who built and those who attend the attractions.

And third, these religious audiences are responding in exactly the way that the research behind science communication would predict, given the way that communication about evolution has commonly been done in the scientific community.

These points reinforce the need for both better evolution education and better science communication. It is on the last point that we'd like to focus: How do scientists better communicate the science behind evolution to audiences that have traditionally struggled to accept it? How do scientists teach evolution in a way that builds bridges of understanding instead of barriers of conflict?

Where has science been?

An effective response to these questions starts by understanding the context of this issue. That is, where has science been, and how did it get to where it is now? Humans have been studying the natural world for millennia and have been documenting said studies as far back as 3000 BCE in ancient Mesopotamia, Egypt, and Babylonia. Much of the more recent history of scientific discovery took place under the umbrella of religious devotion by scientists who saw the natural world as a manifestation of the works of deity, such as Galileo and Mendel (Ferngren 2022). However, around the time that Charles Darwin was publishing his seminal work in England, *On the Origin of Species*, a new movement was beginning that took hold in the United States. With the establishment of Cornell University, one of the first American universities without a religious charter, there was a growing desire by some people to clearly differentiate the secular from the religious. Cornell's founding president, Andrew Dickson White, began a movement to establish religion as an obstacle to intellectual progress, characterized by multiple speaking engagements and culminating in his 1896 publication of *A History of the Warfare of Science with Theology within Christendom* (see Ferngren 2022 for a detailed account). In conjunction with the earlier publication of John William Draper's *History of the Conflict Between Religion and Science* (Draper 1874), which emphasized the repeated impediments that religion made to science throughout history, the ideas that religion and science were fundamentally at odds and that they had been since time immemorial grew quickly within the academy and in broader society.

It was on this cultural stage that the famous Scopes trial took place. The narrative of cultural warfare had never played out more prominently than in this epic "battle" between religious fundamentalism and scientific secularism. Despite the win for the prosecution (although the verdict was later overturned by the Tennessee Supreme Court on a technicality; Dixon and Shapiro 2022), scientists often celebrate it as a win for science in highlighting the controversy over teaching evolution in the public schools, a controversy that has continued to the present day (Plutzer et al. 2020), the latest case being *Kitzmiller v. Dover*, where proponents of intelligent design attempted to make intelligent design a part of the science curriculum; the U.S. District Court sided in favor of the plaintiffs determining intelligent design to be inappropriate for the public school classroom (*Kitzmiller v. Dover* 2004).

Although it is inaccurate, the warfare model of science and religion has become somewhat of a self-fulfilling prophecy, motivating people on both sides of the issue to further the conflict narrative. This is especially apparent when it comes to evolution. In the late nineteenth and earlier twentieth centuries, evangelists like Dwight L. Moody and Billy Sunday argued that Darwinism was a threat to public morality, and, of course, one of the most famous examples was illustrated by William Jennings Bryan, who argued during the Scopes trial that evolution undermined Christian faith (Masci 2009b). More recently, Ken Ham, the founder of *Answers in Genesis*, famously portrayed "man's word" and "God's word" as two dueling castles fighting to obliterate one another, with evolution seemingly being the foundation on which all other conceived social ills rest (Ham 2008). Equally problematic are scientists, such as Richard Dawkins, who treat religion in general as a hostile enemy with statements like, "Faith cannot move mountains… But it… seems to me to qualify as a kind of mental illness" (Dawkins 1989a, p. 198), or physicist Steven Weinberg, who suggests religion leads to evil outcomes: "Religion is an insult to human dignity. With or without religion, you would have good people doing good things and evil people doing evil things. But for good people to do evil things, that takes religion" (Weinberg 1999) At times, it seems that neither side is attempting to communicate in a productive manner. And as Al-Shamahi pointed out in a recent commentary (2025), "When people of faith and political conservatives see their views mocked, dismissed or ostracized, they begin to see science not as a method but as a tribe they're not a part of. And once science becomes just another tribe, its authority collapses."

Where is science now?

Twenty years since the *Kitzmiller v. Dover* trial and 100 years since the Scopes trial, research shows that America still has a large portion of US adults who reject the scientific consensus concerning evolution. In a recent Gallup poll (2024), 37% of the respondents still endorsed a special creation of humans in the last 10,000 years or so. However, a recent Pew poll (Masci 2019) indicates that only 18% truly reject evolution of humans in any form, although that view varies significantly with religious identity. For example, among White evangelical protestants, 38% still reject human evolution, similar to the Gallup poll. But among Catholics, that number is 13%; among White mainline Protestant, it is 16%; and among Black Protestants, it is 27%. Therefore, there is still a large percentage of Americans who reject one of the fundamental principles of biology. What is driving this continued rejection?

First, scientists have sometimes explored a deficit model in attempting to understand attitudes about evolution (e.g., Lawson and Weser 1990, Mead et al. 2017). The assumption is that a lack of some necessary intelligence or scientific reasoning is the causal factor for rejecting evolution, as is captured in this statement by Richard Dawkins, "It is absolutely safe to say that if you meet somebody who claims not to believe in evolution, that person is ignorant, stupid, or insane" (Dawkins 1989b, p. 34). This claim has proven to be unwarranted when it comes to evolution acceptance; research has shown that scientific reasoning is unrelated to evolution acceptance (Manwaring et al. 2018).

Second, there is evidence that a lack of evolutionary knowledge can play at least a part in the rejection of evolutionary theory. Several recent and robust studies show a clear relationship between an understanding of evolution and its acceptance (Glaze et al. 2015, Weisberg et al. 2018, Dunk et al. 2019). Although a more

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consistent predictor of evolution acceptance is an understanding of scientific epistemology, or the nature of science, not an individual's intelligence (Glaze et al. 2015, Dunk et al. 2019).

Third is the influence of worldview. Although a person's worldview is perhaps the most important driver of rejection, it is often more complicated than some might assume. Simple polling shows a clear relationship between certain religious affiliations, especially evangelical Christian, and a rejection of evolutionary theory, most specifically human evolution (Masci 2019), although not all Judeo-Christian affiliations that reject evolution reject it to an equal degree (e.g., Jensen et al. 2019, Lindsay et al. 2019, Masci 2019, Ferguson and Jensen 2021). We have also shown a relationship between more nuanced measures of religiosity (e.g., a common measure uses religious practice, influence, and hope; Sethi and Seligman 1993) and evolution acceptance (see Manwaring et al. 2018). However, recent research suggests that perceived conflict between one's particular religious beliefs and evolution, and not religiosity itself, drives rejection (Barnes et al. 2021a), although this further complicates the narrative. In a nationwide sample, we have shown that the idea of inherent conflict between religion and science is largely being driven by the least religious individuals (Ferguson et al. 2024). Recent Pew polling data confirms this: The least religious respondents were far more likely to acknowledge a conflict (73%) than the most religious respondents (35%; Pew Research Center 2025). Therefore, it is important to emphasize that automatically assuming a religious person will reject evolution is an extreme oversimplification (as is the idea of a broadly stated evolution–religion divide). A more nuanced understanding of the individual's religious identity is needed.

For example, in a recent study using a nationwide survey (Rousseau et al. 2025), we showed a direct relationship between a literal (or "plain") reading of the Bible and evolution denial. This relationship has been confirmed in previous studies as well (Evans 2013, Barnes et al. 2020). And even small decreases in this literalist view, through classroom interventions, can lead to higher evolution acceptance (Rousseau et al. 2025).

Drivers of rejection aside, a large body of research exists regarding ways to address conflicts in worldview and increase acceptance. Often referred to under an umbrella framework, Religious Cultural Competence in Evolution Education (ReCCEE, Barnes and Brownell 2018), there are well-studied mechanisms to reduce conflict among religious students by approaching worldview conflicts with sensitivity and deliberateness. We will highlight a few of these studies: In a 2015 study of college students at a religion-affiliated institution, incorporating a conflict-reducing strategy, referred to as a *r econciliation module*, led to a 13.4% gain in evolution acceptance (Manwaring et al. 2015). Using a similar teaching framework in an introductory biology course showed gains over 30% in acceptance in two cohorts over a 30-year time period (Bradshaw et al. 2018). In a study done at four religion-affiliated institutions of different Christian affiliations, acceptance gains ranged from 18% to 32% when a reconciliatory framework was used prior to evolution instruction (Lindsay et al. 2019). Similarly, a third study run at various institutions across the country (religious and nonreligious) showed that significant gains, especially in human evolution, can be achieved through the use of reconciliatory frameworks (Ferguson and Jensen 2021). Bowen and colleagues (2022) showed an 11% gain in human evolution using an online module that addresses worldview conflicts using a ReCCEE approach. Using these conflict-reducing strategies can even be effective in a theology class; Tolman and colleagues (2020) showed a nearly 4% gain in theology students even when evolutionary content was not covered. We have also tried these approaches outside the traditional biology (or theology) college classroom. Tolman and colleagues (2021) showed that addressing this conflict and offering reconciliation strategies in a simple 15-minute presentation in a public aquarium changed the minds of 40% of the patrons toward a higher acceptance of evolution. Additionally, sophisticated teaching modules have been built for youth pastors and religious teachers that lead to significant gains in acceptance (see Kaloi et al. 2022 and Harrington et al. 2024).

Others have found similar effects. In a large-scale nationwide study, Aini and colleagues (2025) implemented conflict-reducing strategies via video instruction and found that student attitudes about the compatibility of science and religion were significantly increased. Holt and colleagues (2018) found that including a religious role model in evolution instruction significantly predicted a decrease in conflict and increase in acceptance. Bertka and colleagues (2019) showed that conflict around evolutionary theory in high school students was reduced by using a cultural and religious sensitivity module before using human examples of evolution. Barnes and colleagues (2017) showed that a short lesson discussing potential compatibility at a public institution could reduce feelings of conflict by 50% in both religious and nonreligious students. It might be noted that none of these studies have undertaken a longitudinal look at acceptance, so it is unclear how enduring these changes may be. That being said, the immediate changes are indeed encouraging.

As a group of researchers who care deeply about bringing the religious public into the fold of scientific understanding concerning evolution, we have used each of these studies to increase our understanding about the communication strategies that can be truly effective. We will outline these in the next section, incorporating what we learn from science communication research.

Where do scientists go from here?

As was outlined above, since those who reject evolution are responding in the way that science communication research would predict, we would like to explicate the research base underlying successful (and unsuccessful) communication strategies further. We will recommend four strategies for effective communication, grounded in science communication literature—namely, assuming competence and avoiding a deficit mindset, defining the nature of science and acknowledging its limitations, facilitating cognitive consistency, and being culturally sensitive.

Assuming competence and avoiding a deficit mindset

As was discussed above, a deficit model does not hold up to scientific scrutiny (see Manwaring et al. 2018). Researchers have coined the term *social reactance* (see Brehm and Brehm 1981) to describe the phenomenon that occurs when a speaker uses an overly authoritative or condescending tone (i.e., assuming your audience is less intelligent), causing the audience to move further away from where the speaker wants them to be. In other words, social reactance causes the audience to dig in their heels on their current position.

Not only can deficit thinking manifest itself in the assumption that the audience lacks knowledge, but it is often unhelpfully applied to values, morals, or even inherent abilities. For example, if someone denies human-caused climate change, it does not mean that they lack compassion for the environment. If someone believes a COVID vaccine mandate is unconstitutional, it does not mean that they lack the moral understanding that unvacci-

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nated people may hurt others. And as we've shown above, rejecting evolution does not mean that someone lacks scientific reasoning skills.

What does this mean for communication? It means that when scientists discuss evolution with a resistant audience (whether that be college students, the general public, or even outspoken opponents to evolution), we should assume competence and avoid making statements that indicate we assume our audience is deficient in some quality or attribute. Here is some helpful language to try: "There is a lot of conflict surrounding evolution, especially in a religious context, but research clearly shows that both sides are rational and intelligent. Let's look at the data and see if we can reason through it together."

Defining the nature of science and acknowledging its limitations

In the context of evolutionary science, defining the nature and acknowledging the limitations of science has several components: defining our terms, representing science accurately as an agnostic endeavor, and managing the inherent uncertainty in science. We will briefly go through each.

First, defining our terms is essential, especially when it comes to the term *theory*. That evolution is "just a theory" is one of the most common objections used against evolution. For example, several states have attempted to place a disclaimer in biology textbooks stating that because evolution is "theory, not fact," it should be viewed as less robust in some way (see Miller 2000). Clearly, there is a large gap between the common vernacular use of *theory*, as a guess or a hunch, and the scientific definition, which holds much more weight. Other terms that may cause confusion might include *inference*, *statistically significant*, *random*, *natural*, or *mutation* (see Shtulman 2021 for a good review).

What does this mean for communication? Educators should spend time in their classes or conversations making sure that their audience is defining terms the same way as scientists and taking the time to correct misconceptions about terminology in a respectful and constructive way. Here is some helpful language to try: "Scientists label evolution as a *theory*. It's important to understand that the word *theory* in science is much different than the way we use *theory* in common speech. A theory is…" (consider an active learning exercise to teach this concept; see https://biology.byu.edu/reconciling-evolution/recoevo-share-tool, Video #1: How is Science Done?, for an example).

Second, it is important, especially for teaching evolution, that scientists properly represent science as agnostic, or "without knowledge" of a higher or supernatural power. Research has shown that more than half of college students think one must adopt an atheistic viewpoint in order to accept evolution, leading to lower acceptance of the theory (Barnes et al. 2020). As scientists, we should be very careful in the assertions we make concerning what science can and cannot test. Science *can* test the hypothesis that organisms evolve or that organisms share a common ancestor. It cannot test the hypothesis that a god was or was not involved in the process.

What does this mean for communication? Given that the predominant conflict with evolution stems from an individual's religious viewpoints (see Dunk et al. 2019), it means that scientists should avoid claiming that science can disprove religious beliefs outside the purview of science. Whether or not God (or some higher power) exists or played a part in the creation of life on Earth is not a question answered by a scientific epistemology. This likely will go a long way in reducing the identity threat that evolution poses to religious individuals. (We will speak more about identity threats in a later section.) Here is some helpful language to try: "I recognize that many of you have religious beliefs that seem to conflict with what we will learn today. I am going to teach you the science of evolution (this is a science class, after all), but rest assured, science cannot answer whether a creator exists or was involved in the process; we will leave that to religion or philosophy classes."

Third, uncertainty is a necessary part of science and of everyday life for that matter (see Kampourakis and McCain 2019). It is what drives us toward discovery, prompting curiosity and hypothesis testing. However, uncertainty can also be used as a weapon against science when the uncertainty is greatly exaggerated. A helpful distinction to make, taken from physicist Pierre Hohenburg and highlighted in an editorial by Bruce Alberts (Alberts 2012), is between big-S Science and little-s science. Big-S Science is the accumulated knowledge that science has produced. It is what is found in textbooks and taught in science classes. Little-s science, on the other hand, is the process through which scientists discover. It is what is described in scientific articles, popular news pieces, and laboratory notes. Big-S Science is fairly well supported and established (although it can change). Little-s science is messy, often changing, and subject to self-correction through peer-review and repeated experimentation. In big-S Science, it is fairly well accepted that modern diversity came about through evolution and that all life on Earth shares a common ancestor. But the process of little-s science is constantly trying to determine the mechanisms that are most influential in shaping that diversity. Scientists discover new evidence that leads them to different hypotheses all the time. Helping students understand the difference between the two can help prevent scientific impotence discounting, the phenomenon where people interpret the uncertainty to suggest that science is unable to answer these questions (Munro 2010).

What does this mean for communication? It may seem like science communicators are being asked to walk a tightrope, and perhaps that is true. Scientists must help students recognize the limits of our scientific understanding (i.e., where there is uncertainty) while maintaining the integrity of science and its usefulness in answering important questions about the world. Here is some helpful language to try: "What I have presented today represents the scientific consensus on evolution. This means that repeated observations and experimentations have led scientists to accept evolution as the best explanation for modern-day diversity. However, we are constantly discovering new data that help us refine the individual mechanisms involved in the process."

Facilitating cognitive consistency

Humans are deeply uncomfortable with things that do not make sense. This often happens when they encounter something new that does not fit into their current mental schema. This mental discomfort is referred to as *cognitive dissonance*. Instructors purposely use cognitive dissonance to dislodge misconceptions by prompting mental accommodation (Duit et al. 2008). For example, in teaching photosynthesis, instructors will often point out the famous Van Helmont experiment, where he planted a 5-pound willow shoot in 200 pounds of soil, which grew to be a 169-pound willow tree still rooted in 200 pounds of soil (see Hershey 1991 for a good description). This puzzling phenomenon prompts cognitive dissonance in students who assume that plants eat soil, and therefore, the soil should have disappeared. Reasoning downstream of this means they need to rethink what they thought about the world. It opens their minds to new information about

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the basics of carbon capture. The reasoning may look something like this: "I've grown plants in my own home. I know that they need soil to survive and that a seed can turn into a large plant with nothing but water and sunlight. Therefore, reasonably deducing from my observations, plants must gain their mass from the soil. But my instructor is now showing me that the soil remains the same weight even though the tree grew larger. Therefore, my conclusion was wrong: Plants do not eat soil." Naturally, the question follows: Where are they getting their mass? This is a fairly straightforward hole to fill through instruction.

However, what if the cognitive dissonance they experience leads to hazardous downstream consequences that are too conflicting or painful to accept? What if the conclusion they must draw from the new information is hazardous to their identity or values or hazardous to their family relationships? We'll give a hypothetical example of this reasoning: "I've been told by my parents and trusted religious leaders that God created the Earth, animals, and humans in a 7-day period from the dust of the Earth. And although I can see the evidence for gradual adaptations (like antibiotic resistance), I have never seen speciation events occur, and no one was there to document the creation of humans. So since I trust that my parents and religious leaders are wise, I can comfortably accept microevolution while rejecting macro- and human evolution. But my instructor is now showing me compelling evidence that speciation does and has occurred many times and that humans evolved from the same ancestor as chimpanzees. Therefore, my conclusion was wrong: My parents and religious leaders are not wise; in fact, they may have deceived me." This conclusion is a hazardous thought. There are several communication theories that can explain people's reactions to this situation. The first, cognitive dissonance avoidance, has shown that individuals tend to avoid information that disrupts their mental schema unless it is sufficiently convincing (Festinger 1957). The second is motivated cognition (Kunda 1990), which suggests that people selectively (and unconsciously) interpret information in ways that defend their preexisting attitudes. More important, however, is if the cognitively dissonant information is threatening to group membership, such as a religious group, in which case, people are likely to exhibit identity-protective cognition, or the discounting of information that is inconsistent with their in-group because they see it as a threat to their own identity (see Kahan et al. 2007). And interestingly, as their knowledge and cognitive sophistication around the topic increases, identity-protective cognition actually tends to intensify, reinforcing their own views and strengthening their resistance to others' (Drummond and Fischhoff 2017). Certainly, insinuating that their parents or religious leaders are in error would be a threat to their in-group.

What does this mean for communication? It means that, first, scientists must be aware of the cognitive consequences of the information we are teaching. This means scientists must be aware of our audience and their potential cultural conflicts. In the case of evolution, the cultural conflicts and potential consequences of changing a viewpoint have been well studied and can be easily anticipated by instructors (see Heddy and Nadelson 2012 for a good review). It also means that scientists must be careful in our language choices to avoid implying that we think our audience (or people the audience trusts) are unintelligent or purposefully deceitful. Whether the intentions are known or not, implying this will certainly lead to motivated cognition by our audience. Here is some helpful language to try: "I recognize that learning about evolution may bring up some conflict for you with what you have previously learned. I want to assure you that I think you are intelligent and that what you have been previously taught on this subject was done with good intentions. No one is questioning this. I would just like to show you why I and the scientific community have concluded that organisms have evolved."

Being culturally sensitive

This last point is related to the point above. If scientists want to communicate about evolution to audiences who are resistant, we must be culturally sensitive, which is often one of the most difficult challenges when it comes to evolution, especially for scientists. In part, this is because those who teach evolution often do not share an identity with their religious audience. A recent Pew poll (Smith et al. 2025) indicates that 86% of the US population believes in God. The last Pew poll to measure this (Masci 2009a) showed that only 33% of scientists do. Among biologists, that number drops to 27% (Ecklund 2010). Therefore, there is a cultural mismatch between communicator and audience. Recent research has shown that students are especially aware of this, often concealing their Christian identities for fear of stigmatization (see Barnes et al. 2021b for an example with graduate students).

Research clearly shows that stereotype reinforcing or identity- or worldview-attacking language will provoke identity-protective cognition in the audience (Schwarz et al. 2007, Song and Schwarz 2008). In addition, from cognitive load theory (Sweller 2010), the time spent trying to protect their religious identity constitutes extraneous cognitive load, pulling resources away from the cognition they could be using to learn. Researchers have seen this in studies on stereotype threat, where individuals who are worried that they might confirm a stereotype will actually perform more poorly than equally matched peers not under the stereotype threat (Steele and Aronson 1995). In other words, if the audience is feeling like their cultural identity is being attacked, they are likely not listening or comprehending what their scientist teachers are saying. In fact, they may even spend their cognitive energy on constructing alternative arguments and materials to counter evolutionary theory, regardless of factual accuracy, as we have seen our friends on the other side of the aisle spend tremendous amounts of energy doing.

What does this mean for communication? First and foremost, scientists must approach topics in a way that is respectful of others' values and worldviews (see Kahan 2010). We are definitely not implying that scientists should compromise scientific integrity or present anything less than a robust body of research confirming evolutionary theory. Instead, we are suggesting that scientists avoid language that would belittle religious faith or make religious individuals feel that they are unwelcome in the conversation. For example, making the claim that evolution proves that a god does not exist would not only be well beyond the bounds of what science can actually say but also deeply offensive to a religious in-group. Here is some helpful language to try: "All of the scientific evidence seems to support that organisms, including humans, evolved from a common ancestor. That does not mean that God (or a higher power) couldn't have been involved; that's just not a question that science is equipped to answer."

A second strategy that has been well tested (see Holt et al. 2018 and Ferguson and Jensen 2021) is to provide role models from the in-group who embrace the science while maintaining faith. Research has shown that individuals are more likely to assign expert status to those who they believe share their values (Earle and Cvetkovich 1995, Siegrist et al. 2000, Kahan et al. 2010). For evolution, a few exemplary people might be Francis Collins, a former director of the National Institutes of Health; Francisco Ayala, a famous geneticist; Simon Conway Morris, a paleontologist

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at the University of Cambridge, or any member of the Human Origins Initiative Broader Social Impacts Committee of the Smithsonian (for a list of its members and helpful resources for students, see https://humanorigins.si.edu/about/broader-social-impacts-committee).

Conclusions

We close this discussion of the importance of effective communication with a personal message. We are undergraduates currently navigating and professors who once navigated the college experience. We are passionate biologists who are also deeply religious and have had to traverse the divide between science and religion from our first introductory biology classes through our present careers. Since those early beginnings, we have been negatively affected by poor communication from both sides: from well-meaning biologists and science communicators who were unaware of how their dialogue minimized our religious identities and from passionate creationists who discouraged sound scientific thinking and pursuits. At times, the language and tone of each side implied the belief that the two sides were mutually exclusive or, rather, that this was a hopeless divide to cross. The episodes mentioned in this article were consistent with that broader experience. However, both science and religion remain important parts of our identities. We have deeply appreciated the efforts by faculty and colleagues who, perhaps because of a greater awareness of appropriate communication styles, helped us build bridges between our scientific and religious identities. These bridges are more than simple ways to reduce cognitive dissonance; they are crucial tools that allow us to wholly engage in both science and religion without sacrificing aspects of our identity.

We recognize that ours is not a unique experience, and many undergraduate students (and members of the public) urgently need a way to pursue science while maintaining a religious identity. In this increasingly polarized world, a sense of belonging can be just as important to student success as the hours of class they endure (Breakwell 1986, Vignoles et al. 2006). In fact, research shows that the affective experience, often influenced by the way that information is presented, can have a profound impact on the student learning experience (e.g., Fredrickson 2000, Pekrun et al. 2002, Buff et al. 2011, Morgan and D'Mello 2016, and see Tan et al. 2021 for a review). Information presented in a way that evokes a negative emotional response consumes more attentional resources, taking those resources away from learning (Schupp et al. 2007), and according to identity process theory (Breakwell 1986, Vignoles et al. 2006), anything that poses a threat to belonging, self-esteem, or control can have negative effects on learning. If scientists provoke negative emotions by presenting science in a way that seems oppositional to our audience's identity, we may inhibit the audience's ability to problem solve, remember, and participate in higher-order cognitive processing of information (Blair and Razza 2007, Linnenbrink 2007, Rice et al. 2007, Pekrun et al. 2009). The good news is that the scientific community can change the way we present science to be inclusive and conducive to emotions that can positively affect motivation, self-efficacy, academic interest, and academic success (Pekrun et al. 2004, Pekrun et al. 2007). We would propose that this same relationship between affect and understanding exist when communications are directed toward the general public. The main point is that the way scientists communicate science can significantly affect our success in promoting scientific literacy.

Both the educational research and science communication cited in the present article provide evidence for why we claim that our current communication strategies are leading to predictable outcomes among those who oppose evolution. And if scientists fail to adopt better strategies, we are destined to maintain or even widen the gap between science and religion. At this crucial moment, in a highly polarized world, let us pause and consider ways in which we can better spread the good word about science. Let us use the blocks of science to build bridges rather than walls.

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References cited

Kitzmiller v. Dover. 2004. American Civil Liberties Union Pennsylvania (14 December 2004). www.aclupa.org/cases/kitzmiller-v-dover.

Aini RQ, Edwards BA, Summersill A, Epting C, Zheng Y, Brownell SE, Barnes ME. 2025. Evidence for the efficacy of conflict-reducing practices in undergraduate evolution education in a randomized controlled study. *CBE—Life Sciences Education* 24: 27. https://doi.org/10.1187/CBE.24-05-0157.

Alberts B. 2012. The breakthroughs of 2012. *Science* 338: 1511. https://doi.org/10.1126/science.1234108.

Al-Shamahi E. 2025. I'm a former creationist. Here's why "follow the science" failed. *Washington Post* (22 September 2025).

Barnes ME, Brownell SE. 2018. A call to use cultural competence when teaching evolution to religious college students: Introducing religious cultural competence in evolution education (ReCCEE). *CBE—Life Sciences Education* 17: 9. https://doi.org/10.1187/cbe.17-11-0238.

Barnes ME, Dunlop HM, Sinatra GM, Hendrix TM, Zheng Y, Brownell SE. 2020. "Accepting evolution means you can't believe in God": Atheistic perceptions of evolution among college biology students. *CBE—Life Sciences Education* 19: 15. https://doi.org/10.1187/cbe.19-09-0180.

Barnes ME, Elser J, Brownell S. 2017. Impact of a short evolution module on students' perceived conflict between religion and evolution. *American Biology Teacher* 79: 104–111.

Barnes ME, Maas SA, Roberts JA, Brownell SE. 2021. Christianity as a concealable stigmatized identity (CSI) among biology graduate students. *CBE—Life Sciences Education* 20: 9. https://doi.org/10.1187/cbe.20-09-0213

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Barnes ME, Supriya K, Zheng Y, Roberts JA, Brownell SE. 2022. A new measure of students' perceived conflict between evolution and religion (PCoRE) is a stronger predictor of evolution acceptance than understanding or religiosity. *CBE—Life Sciences Education* 21: 42. https://doi.org/10.1187/cbe.21-01-0007.

Bertka CM, Pobiner B, Beardsley P, Watson WA. 2019. Acknowledging students' concerns about evolution: A proactive teaching strategy. *Evolution: Education and Outreach* 12: 3. https://doi.org/10.1186/s12052-019-0095-0.

Blair C, Razza RP. 2007. Relating effortful control, executive function, and false belief understanding to emerging math and literacy ability in kindergarten. *Child Development* 78: 647–663.

Bowen C, Summersill A, Jensen J, Brownell S, Barnes M. 2022. A comparison of online and in-person instruction that includes religious cultural competence. *Journal of Microbiology and Biology Education* 23: e00067–22. https://doi.org/10.1128/jmbe.00067-22.

Boxell L, Gentzkow M, Shapiro JM. 2024. Cross-country trends in affective polarization. *Review of Economics and Statistics* 106: 557–565. https://doi.org/10.1162/rest_a_01160.

Bradshaw WS, Phillips AJ, Bybee SM, Gill RA, Peck SL, Jensen JL. 2018. A longitudinal study of attitudes toward evolution among undergraduates who are members of the Church of Jesus Christ of Latter-Day Saints. *PLOS ONE* 13: e0205798. https://doi.org/10.1371/journal.pone.0205798.

Breakwell GM. 1986. *Coping with Threatened Identities*. Routledge.

Brehm SS, Brehm JW. 1981. *Psychological Reactance: A Theory of Freedom and Control*. Academic Press.

Buff A, Reusser K, Rakoczy K, Pauli C. 2011. Activating positive affective experiences in the classroom: "nice to have" or something more? *Learning and Instruction* 21: 452–466.

Dawkins R. 1989a. *The Selfish Gene*, new ed. Oxford University Press.

Dawkins R. 1989b. Put your money on evolution. *New York Times* 7: 35.

Dixon T, Shapiro AR. 2022. *Science and Religion: A Very Short Introduction*, 2nd ed. Oxford Press.

Draper JW. 1874. *History of the Conflict between Religion and Science*. Appleton. Reprinted by Cambridge University Press, 2009.

Drummond C, Fischhoff B. 2017. Individuals with greater science literacy and education have more polarized beliefs on controversial science topics. *Proceedings of the National Academy of Sciences* 114: 9587–9592. https://doi.org/10.1073/pnas.1704882114.

Duit R, Treagust D, Widodo A. 2008. Teaching science for conceptual change: Theory and practice. Pages 629–646 in Vosniadou S, ed. *International Handbook of Research on Conceptual Change*. Routledge.

Dunk R,, et al. 2019. Evolution education involves a complex landscape of interrelated factors. *Nature Ecology and Evolution* 3: 327–329.

Earle TC, Cvetkovich G. 1995. *Social Trust: Toward a Cosmopolitan Society*. Praeger.

Ecklund EH. 2010. *Science vs. Religion: What Scientists Really Think*. Oxford University Press.

Evans JH. 2013. The growing social and moral conflict between conservative Protestantism and science. *Journal for the Scientific Study of Religion* 52: 368–385. https://doi.org/10.1111/jssr.12022.

Ferguson D, Jensen J. 2021. Role model, compatibility, and knowledge lead to increased evolution acceptance. *Evolution: Education and Outreach* 14: 1. https://doi.org/10.1186/s12052-021-00155-x.

Ferguson DG, Larsen RA, Bailey EG, Jensen JL. 2024. Predicting evolution acceptance among religious students using the predictive factors of evolution acceptance and reconciliation (pFEAR) instrument. *Evolution: Education and Outreach* 17: 7. https://doi.org/10.1186/s12052-024-00215-4.

Ferngren GB. 2022. Science and religion. Pages 200–214 in Kneeland TW, eds. *The Routledge History of American Science*. Routledge.

Festinger L. 1957. *A Theory of Cognitive Dissonance*. Row, Peterson.

Fredrickson BL. 2000. Cultivating positive emotions to optimize health and well-being. *Prevention and Treatment* 3: 1.

Brenan M. 2024. Majority still credits God for humankind, but not Creationism. *Gallup* (22 July 2024). https://news.gallup.com/poll/647594/majority-credits-god-humankind-not-creationism.aspx.

Glaze AL, Goldston MJ, Dantzler J. 2015. Evolution in the Southeastern USA: Factors influencing acceptance and rejection in pre-service science teachers. *International Journal of Science and Mathematics Education* 13: 1189–1209.

Ham K. 2008. Building up the arsenal. *Answers in Genesis* (29 August 2025). https://answersingenesis.org/apologetics/building-up-the-arsenal.

Harrington KM, Nelson HM, Ockey J, Gibson A, Spackman B, Cordero AM, Giberson K, Bybee S, Ferguson D, Jensen JL. 2024. Online professional development course helps secondary educators increase their confidence in teaching evolution to religious audiences. *American Biology Teacher* 86: 78–86. https://doi.org/10.1525/abt.2024.86.2.78.

Heddy BC, Nadelson LS. 2012. A global perspective of the variables associated with acceptance of evolution. *Evolution: Education and Outreach* 5: 412–418. https://doi.org/10.1007/s12052-012-0423-0.

Hershey DR. 1991. Digging deeper into Helmont's famous willow tree experiment. *American Biology Teacher* 53: 458–460. http://www.jstor.org/stable/4449369.

Holt EA, Ogden TH, Durham SL. 2018. The positive effect of role models in evolution instruction. *Evolution: Education and Outreach* 11: 11. https://doi.org/10.1186/s12052-018-0086-6.

Iyengar S, Lelkes Y, Levendusky M, Malhotra N, Westwood SJ. 2019. The origins and consequences of affective polarization in the United States. *Annual Review of Political Science* 22: 129–146. https://doi.org/10.1146/annurev-polisci-051117-073034.

Jensen JL, Manwaring KF, Gill RA, Sudweeks RS, Davies RS, Olsen JA, Phillips AJ, Bybee SM. 2019. Religious affiliation and religiosity and their impact on scientific beliefs in the United States. *BioScience* 69: 292–304.

Kahan D. 2010. Fixing the communication failure. *Nature* 463: 296–297. https://doi.org/10.1038/463296a.

Kahan DM, Braman D, Cohen GL, Gastil J, Slovic P. 2010. Who fears the HPV vaccine, who doesn't, and why? An experimental study of the mechanisms of cultural cognition. *Law and Human Behavior* 34: 501–516. https://doi.org/10.1007/s10979-009-9201-0.

Kahan DM, Braman D, Gastil J, Slovic P, Mertz CK. 2007. Culture and identity-protective cognition: Explaining the white-male effect in risk perception. *Journal of Empirical Legal Studies* 4: 465–505. https://doi.org/10.1111/j.1740-1461.2007.00097.x.

Kaloi M, Hopper JD, Hubble G, Niu ME, Shumway SG, Tolman ER, Jensen JL. 2022. Exploring the relationship between science, religion, and attitudes toward evolution education. *American Biology Teacher* 84: 75–81. https://doi.org/10.1525/abt.2022.84.2.75.

Kampourakis K, McCain K. 2019. *Uncertainty: How It Makes Science Advance*. Oxford University Press.

Kunda Z. 1990. The case for motivated reasoning. *Psychological Bulletin* 108: 480–498. https://doi.org/10.1037/0033-2909.108.3.480.

Lawson AE, Weser J. 1990. The rejection of nonscientific beliefs about life: Effects of instruction and reasoning skills. *Journal of Research in Science Teaching* 27: 589–606.

Lindsay J, Arok A, Bybee SM, Cho W, Cordero AM, Ferguson DG, Jensen JL. 2019. Using a reconciliation model leads to large gains in evolution acceptance. *CBE—Life Sciences Education* 18: 58. https://doi.org/10.1187/cbe.19-04-0080.

Downloaded from https://academic.oup.com/bioscience/advance-article/doi/10.1093/biosci/biaf202/8407672 by University of Nebraska-Omaha user on 17 March 2026

Linnenbrink EA. 2007. The role of affect in student learning: A multidimensional approach to considering the interaction of affect, motivation, and engagement. Pages107–124 in Schutz PA, Pekrun R, eds. *Emotion in Education*. Academic Press.

Manwaring KF, Jensen JL, Gill RA, Bybee SM. 2015. Influencing highly religious undergraduate perceptions of evolution: Mormons as a case study. *Evolution: Education and Outreach* 8: 23. https://doi.org/10.1186/s12052-015-0051-6.

Manwaring KF, Jensen JL, Gill RA, Sudweeks RR, Davies RS, Bybee SM. 2018. Scientific reasoning ability does not predict scientific views on evolution among religious individuals. *Evolution: Education and Outreach* 11: 2.

Mead R, Hejmadi M, Hurst LD. 2017. Scientific aptitude better explains poor responses to teaching of evolution than psychological conflicts. *Nature: Ecology and Evolution* 2: 388–394.

Miller KR. 2000. Dissecting the disclaimer. *Reports of the National Center for Science Education* 20: 30–33.

Morgan B, D'Mello SK. 2016. The influence of positive vs. negative affect on multitasking. *Acta Psychologica* 170: 10–18.

Munro GD. 2010. The scientific impotence excuse: Discounting belief-threatening scientific abstracts. *Journal of Applied Social Psychology* 40: 579–600. https://doi.org/10.1111/j.1559-1816.2010.00588.x.

Pekrun R, Elliot AJ, Maier MA. 2009. Achievement goals and achievement emotions: Testing a model of their joint relations with academic performance. *Journal of Educational Psychology* 101: 115–135.

Pekrun R, Frenzel AC, Goetz T, Perry RP. 2007. The control-value theory of achievement emotions: An integrative approach to emotions in education. Pages 13–36 in Schutz PA, Pekrun R, eds. *Emotion in Education*. Academic Press.

Pekrun R, Goetz T, Perry RP, Kramer K, Hochstadt M, Molfenter S. 2004. Beyond test anxiety: Development and validation of the test emotions questionnaire (TEQ). *Anxiety, Stress, and Coping* 17: 287–316.

Pekrun R, Goetz T, Titz W, Perry RP. 2002. Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative research. *Educational Psychologist* 37: 91–105.

Masci D. 2009a. Scientists and belief. Pew Research Center (5 November 2009). www.pewresearch.org/religion/2009/11/05/scientists-and-belief.

Masci D. 2009b. The social and legal dimensions of the evolution debate in the U.S. Pew Research Center (4 February 2009). www.pewresearch.org/religion/2009/02/04/the-social-and-legal-dimensions-of-the-evolution-debate-in-the-us.

Masci D. 2019. For Darwin Day, 6 facts about the evolution debate. Pew Research Center (11 February 2019). www.pewresearch.org/short-reads/2019/02/11/darwin-day.

Kennedy B, Tyson A. 2023. Americans' trust in scientists, positive views of science continue to decline. Pew Research Center (14 November 2023). www.pewresearch.org/science/2023/11/14/americans-trust-in-scientists-positive-views-of-science-continue-to-decline.

Pew Research Center. 2025. Religious Landscape Study. Pew Research Center. https://doi.org/10.58094/3zs9-jc14.

Plutzer E, Branch G, Reid A. 2020. Teaching evolution in U.S. public schools: A continuing challenge. *Evolution: Education and Outreach* 13: 14. https://doi.org/10.1186/s12052-020-00126-8.

Rice JA, Levine LJ, Pizarro DA. 2007. "Just stop thinking about it": Effects of emotional disengagement on children's memory for educational material. *Emotion* 7: 812–823.

Rousseau G, Bourne D, Harrington K, Abele J, Ferguson D, Jensen J. 2025. Offering alternatives to biblical literalism may be the key to increasing the public's acceptance of evolution. *Journal for the Scientific Study of Religion* 64: 328–340, https://doi.org/10.1111/jssr.70004.

Schupp HT, Stockburger J, Codispoti M, Junghöfer M, Weike AI, Hamm AO. 2007. Selective visual attention to emotion. *Journal of Neuroscience* 27: 1082–1089. 10.1523/JNEUROSCI.3223-06.2007.

Schwarz N, Sanna LJ, Skurnik I, Yoon C. 2007. Metacognitive experiences and the intricacies of setting people straight: Implications for debiasing and public information campaigns. Pages 127–161 in Zanna MP, ed. *Advances in Experimental Social Psychology*, vol. 39. Academic Press. https://doi.org/10.1016/S0065-2601(06)39003-X.

Sethi S, Seligman MEP. 1993. Optimism and fundamentalism. *Psychological Science* 4: 256–259.

Shtulman A. 2021. How intuitive beliefs inoculate us against scientific ones. Pages 89–104 in Jonassen DH, Khine MS, eds. *The Future of Learning: Educational Research in the Age of Digital Transformation*. Springer. https://doi.org/10.1007/978-981-16-4090-0_6.

Siegrist M, Cvetkovich C, Roth C. 2000. Salient value similarity, social trust, and risk/benefit perception. *Risk Analysis* 20: 353–362. https://doi.org/10.1111/0272-4332.203034.

Smith GA, Cooperman A, Mohamed B, Rotolo C, Tevington P, Nortey J, Kallo A, Diamant J, Fahmy D. 2025. Decline of Christianity in the U.S. has slowed, may have leveled off: Executive summary. Pew Research Center (26 February 2025). www.pewresearch.org/religion/2025/02/26/religious-landscape-study-executive-summary.

Song H, Schwarz N. 2008. Fluency and the detection of distortions: Low processing fluency attenuates the Moses illusion. *Social Cognition* 26: 791–799. https://doi.org/10.1521/soco.2008.26.6.791.

Stanovich KE. 2021. *The Bias that Divides Us: The Science and Politics of Myside Thinking*. MIT Press.

Steele CM, Aronson J. 1995. Stereotype threat and the intellectual test performance of African Americans. *Journal of Personality and Social Psychology* 69: 797–811. https://doi.org/10.1037/0022-3514.69.5.797.

Sweller J. 2010. Element interactivity and intrinsic, extraneous, and germane cognitive load. *Educational Psychology Review* 22: 123–138. https://doi.org/10.1007/s10648-010-9128-5.

Tan J, Mao J, Jiang Y, Gao M. 2021. The influence of academic emotions on learning effects: A systematic review. *International Journal of Environmental Research and Public Health* 18: 9678. https://doi.org/10.3390/ijerph18189678.

Tolman ER, Ferguson DG, Mann M, Cordero AM, Jensen JL. 2020. Reconciling evolution: Evidence from a biology and theology course. *Evolution: Education and Outreach* 13: 19. https://doi.org/10.1186/s12052-020-00134-5.

Tolman ER, Shumway S, Williams L, Jensen JL. 2021. Testing the effect of aquarium-based learning on patron acceptance of evolutionary theory. *Journal of Zoo and Aquarium Research* 9: 102–109.

Vignoles VL, Regalia C, Manzi C, Golledge J, Scabini E. 2006. Beyond self-esteem: Influence of multiple motives on identity construction. *Journal of Personality and Social Psychology* 90: 308–333.

Weinberg S. 1999. *Address at the Conference on Cosmic Design*. American Association for the Advancement of Science.

Weisberg DS, Landrum AR, Metz SE, Weisberg M. 2018. No missing link: Knowledge predicts acceptance of evolution in the United States. *BioScience* 68: 212–222.

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